

Homework #7

Kevin Fang

June 6, 2025

Question 3.

(a): 8.2.2

B.

For $T(n) = n^3 + 3n^2 + 4$ and $h(n) = n^3$, $n^3 + 3n^2 + 4 = \Theta(n^3)$ if and only if $T(n) = O(h(n))$ and $T(n) = \Theta(h(n))$.

For $n \geq 1$, $n \leq n^2 \leq n^3$.

$$\begin{aligned}n^3 + 3n^2 + 4 &\leq n^3 + 3n^3 + 4n^3 \\n^3 + 3n^2 + 4 &\leq 8n^3 \\n^3 + 3n^2 + 4 &\leq 8h(n)\end{aligned}$$

$T(n) = O(h(n))$ since there are positive constants $c = 8$ and $n_0 = 1$ such that $T(n) \leq ch(n)$ when $n \geq n_0$.

For $n \geq 1$, $3n^2 + 4 \geq 0$

$$\begin{aligned}3n^2 + 4 &\geq 0 \\n^3 + 3n^2 + 4 &\geq n^3 \\n^3 + 3n^2 + 4 &\geq h(n)\end{aligned}$$

$T(n) = \Omega(h(n))$ since there are positive constants $c = 1$ and $n_0 = 1$ such that $T(n) \geq ch(n)$ when $n \geq n_0$.
 $n^3 + 3n^2 + 4 = \Theta(n^3)$ since $T(n) = O(h(n))$ and $T(n) = \Omega(h(n))$.

(b): 8.3.5

A.

This is a quick sort algorithm. The algorithm partitions the array around the pivot p . Specifically, it rearranges the array such that:

- All elements less than p are moved to the left side of the array.
- All elements greater than or equal to p are moved to the right side of the array.

For example, if the input array is $[3, -1, 2, 0, -2, 5]$ and $p = 0$, the output will be $[-2, -1, 2, 0, 3, 5]$ or any other permutation where all elements less than 0 are on the left and all elements greater than or equal to 0 are on the right.

B.

The total number of times the lines $i := i + 1$ or $j := j - 1$ are executed depends on the distribution of numbers relative to p . Specifically:

- Maximizing Executions: If all elements are less than p , i will be incremented $n - 1$ times, and j will not be decremented at all.
- Minimizing Executions: If all elements are greater than or equal to p , j will be decremented $n - 1$ times, and i will not be incremented at all.

The answer of the number of times the lines $i := i + 1$ or $j := j - 1$ are executed depends on the actual values of the numbers in the sequence, not just the length of the sequence. However, the total number of times the lines $i := i + 1$ or $j := j - 1$ are executed depends on the length of the sequence.

C.

The total number of times the swap operation is executed depends on how many elements are on the wrong side of p :

- Maximizing Swaps: If half of the elements are less than p and half are greater than or equal to p , swaps will be maximized to $\lceil \frac{n}{2} \rceil$.
- Minimizing Swaps: If all elements are either less than p or greater than or equal to p , no swaps will occur.

The number of swaps depends on the actual values of the numbers in the sequence, not just the length of the sequence.

D.

The time complexity will be $\Omega(n)$ despite whether it is the worst-case or not.

E.

The time complexity will be $O(n)$.

Question 4.

(a): 5.1.2

B.

$$40^9 + 40^8 + 40^7$$

C.

$$14 * (40^8 + 40^7 + 40^6)$$

(b): 5.3.2

A.

$$2^9 * 3$$

(c): 5.3.3

B.

$$10 * 9 * 8 * 26^4$$

C.

$$10 * 9 * 8 * 26 * 25 * 24 * 23$$

(d): 5.2.3

A.

We define a function $f : B^9 \rightarrow E_{10}$ by adding a 0 to the end of the string if it has even number of 1's and add a 1 to the end of the string if it has odd number of 1's. The function is a bijection if since it has a well defined inverse $f^{-1} : E_{10} \rightarrow B^9$ that can be retrieved by removing the last bit.

B.

$$|E_{10}| = |B_9| = 2^9 = 512$$

Question 5.

(a): 5.4.2

a.

$$10^4 * 2 = 20000$$

b.

$$10 * 9 * 8 * 7 * 2 = 10080$$

(b): 5.5.3

a.

$$2^{10} = 1024$$

b.

$$2^7 = 128$$

c.

$$2^7 + 2^8 = 384$$

d.

$$2^8 = 256$$

e.

$$\binom{10}{6} = 210$$

f.

$$\binom{9}{6} = 84$$

g.

$$\binom{5}{1} \binom{5}{3} = 50$$

(c): 5.5.5

a.

$$\binom{30}{10} \binom{35}{10} = 5515645706510940$$

(d): 5.5.8

c.

$$\binom{26}{5} = 65780$$

d.

$$\binom{13}{1} * \binom{48}{1} = 624$$

e.

$$13 * 12 * \binom{4}{2} * \binom{4}{3} = 3744$$

f.

$$\binom{13}{5} * 4^5 = 1317888$$

(e): 5.6.6

a.

$$\binom{44}{5}\binom{56}{5} = 4148350734528$$

b.

$$P(44, 2) * P(56, 2) = 5827360$$

Question 6.

(a): 5.7.2

A.

$$\binom{52}{5} - \binom{39}{5} = 2023203$$

B.

$$\binom{52}{5} - \left(\binom{13}{5} * 4^5\right) = 1281072$$

(b): 5.8.4

A.

$$5^{20} = 95367431640625$$

B.

$$\binom{20}{4}\binom{16}{4}\binom{12}{4}\binom{8}{4}\binom{4}{4} = 305540235000$$

Question 7.

(a):

$$0$$

(b):

$$P(5, 5) = 120$$

(c):

$$P(6, 5) = 720$$

(d):

$$P(7, 5) = 2520$$